

Operations Management

Complete Formulas & Numerical Problems with Solutions

MMS Examination Preparation Guide

1 Plant Location - Formulas & Numericals

Break-Even Analysis

$$\text{BEP (units)} = F / (S - V)$$

Where:

F = Fixed Cost

S = Selling Price per unit

V = Variable Cost per unit

(S - V) = Contribution per unit

Total Cost at a Location

$$TC = \text{Fixed Cost} + (\text{Variable Cost per unit} \times \text{Quantity})$$

$$TC = F + V(Q)$$

Crossover Point (Between Two Locations)

$$F_A + V_A(X) = F_B + V_B(X)$$

Solve for X to find the production volume at which both locations have equal total cost.

Decision Rule: Below crossover - choose location with lower fixed cost; Above crossover - choose location with lower variable cost.

Problem 1 Break-Even Location Analysis**Problem Statement:**

A company is considering two locations for a new plant:

- Location X: Fixed Cost = Rs. 150,000, Variable Cost = Rs. 10/unit
- Location Y: Fixed Cost = Rs. 350,000, Variable Cost = Rs. 8/unit

Determine which location is appropriate if expected demand is 130,000 units.

Given Data:

Parameter	Location X	Location Y
Fixed Cost (F)	Rs. 150,000	Rs. 350,000
Variable Cost (V)	Rs. 10/unit	Rs. 8/unit

Solution:**1****Find Crossover Point**

Set total costs equal:

$$F_X + V_X(Q) = F_Y + V_Y(Q)$$
$$150,000 + 10Q = 350,000 + 8Q$$

2 Solve for Q

$$10Q - 8Q = 350,000 - 150,000$$
$$2Q = 200,000$$
$$Q = 100,000 \text{ units}$$

3 Decision Rule

If demand < 100,000: Choose Location X (lower fixed cost)
If demand > 100,000: Choose Location Y (lower variable cost)

Given demand = 130,000 units > 100,000 units

4 Verification

TC at Location X = $150,000 + 10(130,000) = \text{Rs. } 1,450,000$
TC at Location Y = $350,000 + 8(130,000) = \text{Rs. } 1,390,000$

Answer

Location Y is appropriate (Saves Rs. 60,000)

Centre of Gravity Method

$$C_x = (\text{Sum of } d_{ix} \times W_i) / (\text{Sum of } W_i)$$

$$C_y = (\text{Sum of } d_{iy} \times W_i) / (\text{Sum of } W_i)$$

Where:

C_x, C_y = Coordinates of centre of gravity

d_{ix}, d_{iy} = X and Y coordinates of location i

W_i = Weight or volume shipped to/from location i

Problem 2 Centre of Gravity Method

Problem Statement:

A company needs to locate a warehouse to serve 4 retail stores. Find the optimal location using Centre of Gravity method.

Given Data:

Store	X-Coordinate	Y-Coordinate	Weight (tons/month)
A	100	200	500
B	300	100	300
C	400	300	200
D	200	400	100

Solution:

- 1
- Calculate Sum of Weights

$$\text{Sum } W_i = 500 + 300 + 200 + 100 = 1,100 \text{ tons}$$

2 Calculate C_x (X-coordinate)

$$\begin{aligned}\text{Sum}(d_{ix} \times W_i) &= (100 \times 500) + (300 \times 300) + (400 \times 200) + (200 \times 100) \\ &= 50,000 + 90,000 + 80,000 + 20,000 = 240,000\end{aligned}$$

$$C_x = 240,000 / 1,100 = 218.18$$

3 Calculate C_y (Y-coordinate)

$$\begin{aligned}\text{Sum}(d_{iy} \times W_i) &= (200 \times 500) + (100 \times 300) + (300 \times 200) + (400 \times 100) \\ &= 100,000 + 30,000 + 60,000 + 40,000 = 230,000\end{aligned}$$

$$C_y = 230,000 / 1,100 = 209.09$$

Answer

Optimal Location: (218.18, 209.09) or approximately (218, 209)

Weighted Factor Rating Method

Total Score = Sum of (Weight_i x Score_i) for all factors

Steps:

1. Identify important factors
2. Assign weights to each factor (total = 1.0 or 100)
3. Score each location for each factor (e.g., 1-100)
4. Calculate weighted score = Weight x Score
5. Sum weighted scores; Select location with highest total

2 Work Study - Formulas & Numericals

Basic Time Calculation

$$\text{Basic Time} = \text{Observed Time} \times (\text{Rating} / 100)$$

Where:

Observed Time (OT) = Time recorded using stopwatch

Rating = Performance rating (100 = standard pace, >100 = faster, <100 = slower)

Standard Time Calculation

$$\text{Standard Time} = \text{Basic Time} + \text{Allowances}$$

OR

$$\text{Standard Time} = \text{Basic Time} \times (1 + \text{Allowance \%})$$

Types of Allowances:

- Personal Allowance (5-7%)
- Fatigue Allowance (4-10%)
- Contingency Allowance (2-5%)
- Special Allowance (startup, shutdown, cleaning)
- Rework Allowance

Normal Time (Alternative Term)

$$\text{Normal Time} = \text{Observed Time} \times (\text{Rating} / 100)$$

Normal Time and Basic Time are used interchangeably in time study calculations.

Problem 1 Basic Time Study Calculation**Problem Statement:**

In a time study, observed time is 80 seconds and performance rating is 80%. Calculate the standard time if total allowance is 15%.

Given Data:

- Observed Time (OT) = 80 seconds
- Performance Rating = 80%
- Total Allowance = 15%

Solution:**1****Calculate Basic Time**

$$\text{Basic Time} = \text{Observed Time} \times (\text{Rating} / 100)$$

$$\text{Basic Time} = 80 \times (80 / 100)$$

$$\text{Basic Time} = 80 \times 0.80 = 64 \text{ seconds}$$

2**Calculate Standard Time**

Standard Time = Basic Time $\times$ (1 + Allowance %)

Standard Time = $64 \times (1 + 0.15)$

Standard Time = $64 \times 1.15 = 73.6$ seconds

Answer

Basic Time = 64 seconds | Standard Time = 73.6 seconds

Problem 2 Multi-Element Time Study (From PPT)

Problem Statement:

The observed times and performance ratings for 5 elements of a job are given. Calculate standard time assuming rest and personal allowance as 15% and contingency allowance as 2% of basic time.

Given Data:

Element	Observed Time (Min)	Performance Rating
1	0.20	85%
2	0.08	80%
3	0.50	90%
4	0.12	85%
5	0.10	80%

Rest & Personal Allowance = 15%, Contingency Allowance = 2%

Solution:**1 Calculate Basic Time for Each Element**

Element	OT (Min)	Rating	Basic Time = OT x (R/100)
1	0.20	85%	$0.20 \times 0.85 = 0.170$
2	0.08	80%	$0.08 \times 0.80 = 0.064$
3	0.50	90%	$0.50 \times 0.90 = 0.450$
4	0.12	85%	$0.12 \times 0.85 = 0.102$
5	0.10	80%	$0.10 \times 0.80 = 0.080$
Total Basic Time			0.866 min

2 Calculate Total Allowance

Total Allowance = Rest & Personal + Contingency

Total Allowance = 15% + 2% = 17%

3 Calculate Standard Time

Standard Time = Basic Time x (1 + Allowance %)

Standard Time = 0.866 x (1 + 0.17)

Standard Time = 0.866 x 1.17 = 1.013 minutes

Answer

Total Basic Time = 0.866 min | Standard Time = 1.013 min (60.78 seconds)

Problem 3 Time Study with Variable Relaxation Allowances (From PPT)**Problem Statement:**

Following are the elements of a machining operation. Calculate normal time and standard time for this job. Assume contingency allowance to be 3%.

Given Data:

Sr. No.	Observed Time (Min)	Rating %	Relaxation Allowance %
1	1.80	60%	10%
2	2.20	90%	12%
3	6.30	100%	10%
4	3.20	100%	10%
5	1.40	90%	12%

Contingency Allowance = 3%

Solution:

1 Calculate Normal/Basic Time and Standard Time for Each Element

Element	OT	Rating	Normal Time	Relax. Allow.	Contingency	Total Allow.	Std Time
1	1.80	60%	$1.80 \times 0.60 = 1.08$	10%	3%	13%	$1.08 \times 1.13 = 1.22$
2	2.20	90%	$2.20 \times 0.90 = 1.98$	12%	3%	15%	$1.98 \times 1.15 = 2.28$
3	6.30	100%	$6.30 \times 1.00 = 6.30$	10%	3%	13%	$6.30 \times 1.13 = 7.12$
4	3.20	100%	$3.20 \times 1.00 = 3.20$	10%	3%	13%	$3.20 \times 1.13 = 3.62$
5	1.40	90%	$1.40 \times 0.90 = 1.26$	12%	3%	15%	$1.26 \times 1.15 = 1.45$
Total			13.82 min				15.69 min

Answer

Normal Time = 13.82 minutes | Standard Time = 15.69 minutes

3 Value Engineering - Formulas

Value Equation

$$\text{Value} = \text{Function} / \text{Cost}$$

Ways to Improve Value:

1. Same function at lower cost (Cost reduction)
2. Better function at same cost (Performance improvement)
3. Better function at lower cost (Ideal case)
4. Much better function at slightly higher cost
5. Slightly lower function at much lower cost

Value Index

$$\text{Value Index} = \text{Function Worth} / \text{Function Cost}$$

Interpretation:

- Value Index < 1: Poor value, needs improvement
- Value Index = 1: Fair value
- Value Index > 1: Good value

Value Engineering Job Plan Steps

- 1 Information Phase:** Gather data - What is it? What does it do? What does it cost?
- 2 Analysis Phase:** Function analysis - What should it do? What is the function worth?
- 3 Creative Phase:** Brainstorming - What else can perform the function?
- 4 Evaluation Phase:** Screen ideas - Which ideas are feasible?
- 5 Development Phase:** Develop solutions - What will it cost? Will it work?
- 6 Presentation Phase:** Present recommendations with cost-benefit analysis

7

Implementation Phase: Implement approved changes

4 Inventory Control & EOQ - Formulas & Numericals

Economic Order Quantity (EOQ)

$$EOQ = \sqrt{(2 \times D \times C_o / C_c)}$$

Where:

D = Annual Demand (units/year)

C_o = Ordering Cost per order (Rs./order)

C_c = Annual Carrying/Holding Cost per unit (Rs./unit/year)

Alternative: If C_c is given as percentage of unit cost:

C_c = Carrying Cost % x Unit Cost

Total Inventory Cost

$$\text{Total Cost} = \text{Ordering Cost} + \text{Carrying Cost}$$

$$TC = (C_o \times D/Q) + (C_c \times Q/2)$$

Where:

D/Q = Number of orders per year

$Q/2$ = Average inventory level

At EOQ: Ordering Cost = Carrying Cost (costs are balanced)

Related EOQ Calculations

$$\text{Number of Orders per Year} = D / \text{EOQ}$$

$$\text{Time Between Orders} = \text{EOQ} / D \text{ (in years)}$$

$$\text{Reorder Point} = \text{Demand during Lead Time} = d \times L$$

Where: d = daily demand, L = lead time in days

Problem 1 Basic EOQ Calculation**Problem Statement:**

A company has annual demand of 10,000 units. Ordering cost is Rs. 50 per order and carrying cost is Rs. 2 per unit per year. Calculate EOQ, number of orders, and total inventory cost.

Given Data:

- Annual Demand (D) = 10,000 units
- Ordering Cost (Co) = Rs. 50 per order
- Carrying Cost (Cc) = Rs. 2 per unit per year

Solution:**1 Calculate EOQ**

$$\begin{aligned} \text{EOQ} &= \sqrt{(2 \times D \times C_o / C_c)} \\ \text{EOQ} &= \sqrt{(2 \times 10,000 \times 50 / 2)} \\ \text{EOQ} &= \sqrt{(1,000,000 / 2)} \\ \text{EOQ} &= \sqrt{500,000} \\ \text{EOQ} &= 707.11 \approx 707 \text{ units} \end{aligned}$$

2 Calculate Number of Orders per Year

Number of Orders = D / EOQ

Number of Orders = $10,000 / 707$

Number of Orders = $14.14 \approx 14$ orders/year

3

Calculate Total Inventory Cost

Total Cost = $(Co \times D/Q) + (Cc \times Q/2)$

Total Cost = $(50 \times 10,000/707) + (2 \times 707/2)$

Total Cost = $(50 \times 14.14) + (2 \times 353.5)$

Total Cost = $707 + 707$

Total Cost = Rs. 1,414

Answer

EOQ = 707 units | Orders/Year = 14 | Total Cost = Rs. 1,414

Problem 2 EOQ with Storage Constraint (From PPT)**Problem Statement:**

Ashish keeps his inventory in special containers. Each container occupies 10 sq.ft. of store space. Only 5000 sq.ft. of storage space is available. The annual demand for the inventory item is 9000 containers, priced at Rs. 8 per container. The ordering cost is Rs. 40 per order, and the annual carrying costs amount to 25% of inventory value. Would you recommend to Ashish to increase his storage space? If so, how much should be the increase?

Given Data:

- Available Storage = 5000 sq.ft.
- Space per container = 10 sq.ft.
- Maximum storage capacity = $5000/10 = 500$ containers
- Annual Demand (D) = 9000 containers
- Unit Price (P) = Rs. 8
- Ordering Cost (Co) = Rs. 40/order
- Carrying Cost (Cc) = 25% of Rs. 8 = Rs. 2/unit/year

Solution:**1****Calculate EOQ**

$$C_c = 25\% \times 8 = \text{Rs. } 2 \text{ per unit per year}$$

$$EOQ = \sqrt{(2 \times D \times C_o / C_c)}$$

$$EOQ = \sqrt{(2 \times 9000 \times 40 / 2)}$$

$$EOQ = \sqrt{(720,000 / 2)}$$

$$EOQ = \sqrt{360,000}$$

$$EOQ = 600 \text{ containers}$$

2 Compare with Storage Capacity

$$EOQ = 600 \text{ containers}$$

$$\text{Current Max Storage} = 500 \text{ containers}$$

$$\text{Shortfall} = 600 - 500 = 100 \text{ containers}$$

3 Calculate Additional Space Required

$$\text{Additional containers} = 100$$

$$\text{Space per container} = 10 \text{ sq.ft.}$$

$$\text{Additional space needed} = 100 \times 10 = 1000 \text{ sq.ft.}$$

4 Cost Comparison

At Current Capacity (Q = 500):

$$TC = (40 \times 9000/500) + (2 \times 500/2) = 720 + 500 = \text{Rs. } 1,220$$

At EOQ (Q = 600):

$$TC = (40 \times 9000/600) + (2 \times 600/2) = 600 + 600 = \text{Rs. } 1,200$$

$$\text{Savings} = 1220 - 1200 = \text{Rs. } 20/\text{year}$$

Answer

Yes, increase storage by 1000 sq.ft. (to store 600 containers)

Annual savings = Rs. 20

Problem 3 EOQ with Quantity Discount (From PPT)**Problem Statement:**

GSB Department store stocks toy cars. The store was given a quantity discount schedule. Normal cost is \$5.00; for orders 1000-1999 units, unit cost is \$4.80; for orders 2000+ units, cost is \$4.75. Ordering cost is \$49 per order, annual demand is 5,000 cars, and carrying cost is 20% of unit cost. Find optimal order quantity.

Given Data:

Order Quantity	Unit Price
1-999	\$5.00
1000-1999	\$4.80
2000+	\$4.75

- $D = 5,000$ units/year
- $C_o = \$49/\text{order}$
- Carrying cost = 20% of unit price

Solution:

1

Calculate EOQ for Each Price Level

Price \$5.00: $C_c = 0.20 \times 5.00 = \1.00

$EOQ = \sqrt{(2 \times 5000 \times 49 / 1.00)} = \sqrt{490,000} = 700$ units

(Valid: 700 is in range 1-999) ✓

Price \$4.80: $C_c = 0.20 \times 4.80 = \0.96

$EOQ = \sqrt{(2 \times 5000 \times 49 / 0.96)} = \sqrt{510,417} = 714$ units

(Invalid: 714 not in range 1000-1999, adjust to 1000) ✗

Price \$4.75: $C_c = 0.20 \times 4.75 = \0.95

$EOQ = \sqrt{(2 \times 5000 \times 49 / 0.95)} = \sqrt{515,789} = 718$ units

(Invalid: 718 not in range 2000+, adjust to 2000) ✗

2

Calculate Total Cost for Valid Options

TC = Purchase Cost + Ordering Cost + Carrying Cost

$TC = P \times D + C_o \times D/Q + C_c \times Q/2$

Q = 700, P = \$5.00:

$TC = (5.00 \times 5000) + (49 \times 5000/700) + (1.00 \times 700/2)$

$TC = 25,000 + 350 + 350 = \$25,700$

Q = 1000, P = \$4.80:

$TC = (4.80 \times 5000) + (49 \times 5000/1000) + (0.96 \times 1000/2)$

$TC = 24,000 + 245 + 480 = \$24,725$

Q = 2000, P = \$4.75:

$$TC = (4.75 \times 5000) + (49 \times 5000/2000) + (0.95 \times 2000/2)$$

$$TC = 23,750 + 122.50 + 950 = \$24,822.50$$

Answer

Optimal Order Quantity = 1,000 units at \$4.80 each

Minimum Total Cost = \$24,725

ABC Analysis Calculation

Annual Usage Value = Unit Cost x Annual Usage

Steps:

1. Calculate annual usage value for each item
2. Rank items in descending order of usage value
3. Calculate cumulative percentage of value
4. Classify: First ~70-80% value = A, Next ~15% = B, Remaining = C

Problem 4 ABC Analysis (From PPT)**Problem Statement:**

Perform ABC analysis on the following inventory items:

Given Data:

Item	Annual Consumption	Price/Unit (Rs.)
1	300	0.10
2	2800	0.15
3	30	0.10
4	1100	0.05
5	40	0.05
6	220	1.00
7	150	0.05
8	800	0.05

9	600	0.15
10	80	0.10

Solution:

1 Calculate Annual Usage Value and Rank

Item	Consumption	Price	Usage Value	Rank
1	300	0.10	30	6
2	2800	0.15	420	1
3	30	0.10	3	10
4	1100	0.05	55	4
5	40	0.05	2	9
6	220	1.00	220	2
7	150	0.05	7.5	8
8	800	0.05	40	5

9	600	0.15	90	3
10	80	0.10	8	7

2 Arrange by Rank and Calculate Cumulative %

Rank	Item	Usage Value	% of Total	Cumulative %	Class
1	2	420	47.9%	47.9%	A
2	6	220	25.1%	73.0%	
3	9	90	10.3%	83.3%	B
4	4	55	6.3%	89.6%	
5	8	40	4.6%	94.2%	C
6	1	30	3.4%	97.6%	
7	10	8	0.9%	98.5%	
8	7	7.5	0.9%	99.4%	
9	3	3	0.3%	99.7%	
10	5	2	0.2%	100%	

Total	875.5	100%		
-------	-------	------	--	--

Answer

Class A: Items 2, 6 (20% of items, 73% of value)

Class B: Items 9, 4 (20% of items, 16.6% of value)

Class C: Items 8, 1, 10, 7, 3, 5 (60% of items, 10.4% of value)

5 Sequencing - Methods & Numericals

Johnson's Rule (n Jobs, 2 Machines in sequence AB)

- 1 List all jobs with processing times on Machine A and Machine B
- 2 Find the minimum processing time among all jobs on both machines
- 3 If minimum is on Machine A → Place job FIRST in sequence
If minimum is on Machine B → Place job LAST in sequence
- 4 Remove the job from list and repeat until all jobs are sequenced
- 5 Create timeline (Gantt chart) and calculate total elapsed time

Problem 1 Johnson's Rule - 5 Jobs, 2 Machines (From PPT)**Problem Statement:**

There are 5 jobs, each of which has to go through machines A and B in the order AB. Determine the optimal sequence to minimize total elapsed time. Also find idle time for each machine.

Given Data:

Job	J1	J2	J3	J4	J5
Machine A	2	4	5	7	1
Machine B	3	6	1	4	8

Solution:**1****Apply Johnson's Rule**

Iteration 1: Min = 1 (J5 on A and J3 on B)

J5 has min on A → Place J5 FIRST

Sequence: [J5, __, __, __, __]

Iteration 2: Min = 1 (J3 on B) among remaining

J3 has min on B → Place J3 LAST

Sequence: [J5, _, _, _, J3]

Iteration 3: Min = 2 (J1 on A) among remaining

J1 has min on A → Place J1 next from front

Sequence: [J5, J1, _, _, J3]

Iteration 4: Min = 4 (J2 on A or J4 on B)

J2 has min on A → Place J2 next from front

Sequence: [J5, J1, J2, _, J3]

Iteration 5: Only J4 remains

Sequence: [J5, J1, J2, J4, J3]

2 Calculate Time Schedule

Sequence	Job	Machine A		Machine B	
		In	Out	In	Out
1	J5	0	1	1	9
2	J1	1	3	9	12
3	J2	3	7	12	18

4	J4	7	14	18	22
5	J3	14	19	22	23

3

Calculate Idle Times

Machine A: Idle at end = $23 - 19 = 4$ hours
 (Machine A finishes at 19, total time is 23)

Machine B: Idle at start = 1 hour (waits for J5 from A)

B also has idle between jobs where it waits for A

Machine B idle = $1 + (9-9) + (12-12) + (18-18) + (22-22) = 1$ hour initially

Plus gaps: Check each job transition...

Answer

Optimal Sequence: J5 → J1 → J2 → J4 → J3

Total Elapsed Time = 23 hours

Machine A Idle Time = 4 hours | Machine B Idle Time = 1 hour

Extended Johnson's Rule (n Jobs, 3 Machines in sequence ABC)

Applicability Condition:
$$\text{Min}(A) \geq \text{Max}(B) \text{ OR } \text{Min}(C) \geq \text{Max}(B)$$
Conversion Method:

- 1 Check if conditions are satisfied
- 2 Create virtual machines: $G = A + B$, $H = B + C$
- 3 Apply standard Johnson's rule on G and H
- 4 Use the resulting sequence for all three machines

Problem 2 Extended Johnson's Rule - 3 Machines (From PPT)**Problem Statement:**

Find the sequence that minimizes total time for 5 jobs on 3 machines in order ABC.

Given Data:

Job	1	2	3	4	5
Machine A	8	10	6	7	11
Machine B	5	6	2	3	4
Machine C	4	9	8	6	5

Solution:**1****Check Applicability Condition**

$$\text{Min}(A) = \min(8, 10, 6, 7, 11) = 6$$

$$\text{Max}(B) = \max(5, 6, 2, 3, 4) = 6$$

$$\text{Min}(C) = \min(4, 9, 8, 6, 5) = 4$$

Condition 1: $\text{Min}(A) \geq \text{Max}(B) \rightarrow 6 \geq 6 \checkmark$ (Satisfied)

Extended Johnson's Rule can be applied.

2 Create Virtual Machines G and H

Job	1	2	3	4	5
G = A + B	$8+5=13$	$10+6=16$	$6+2=8$	$7+3=10$	$11+4=15$
H = B + C	$5+4=9$	$6+9=15$	$2+8=10$	$3+6=9$	$4+5=9$

3 Apply Johnson's Rule on G and H

Iteration 1: $\text{Min} = 8$ (Job 3 on G) $\rightarrow$ J3 FIRST

Sequence: [J3, __, __, __, __]

Iteration 2: $\text{Min} = 9$ (Jobs 1, 4, 5 on H)

Pick any with min on H $\rightarrow$ J1 LAST (or J4 or J5)

Let's use J1 $\rightarrow$ Sequence: [J3, __, __, __, J1]

Iteration 3: $\text{Min} = 9$ (J4 or J5 on H)

J4 $\rightarrow$ LAST available $\rightarrow$ Sequence: [J3, __, __, J4, J1]

Iteration 4: $\text{Min} = 9$ (J5 on H)

J5 → LAST available → Sequence: [J3, _, J5, J4, J1]

Iteration 5: Only J2 remains

Sequence: [J3, J2, J5, J4, J1]

Answer

Optimal Sequence: J3 → J2 → J5 → J4 → J1

Problem 3 Processing 5 Jobs Through 4 Machines (From PPT)**Problem Statement:**

Find the solution for processing 5 jobs through 4 machines in sequence.

Given Data:

Job	1	2	3	4	5
Machine-1	11	13	9	16	17
Machine-2	4	3	5	2	6
Machine-3	6	7	5	8	4
Machine-4	15	8	13	9	11

Solution:**1****Check Conditions for 4-Machine Problem**

$\text{Min}(M1) = 9, \text{Min}(M4) = 8$

$\text{Max}(M2) = 6, \text{Max}(M3) = 8$

Condition: $\text{Min}(M1) \geq \text{Max}(M2)$ AND $\text{Min}(M1) \geq \text{Max}(M3)$

OR $\text{Min}(M4) \geq \text{Max}(M2)$ AND $\text{Min}(M4) \geq \text{Max}(M3)$

Check: $9 \geq 6 \checkmark$ AND $9 \geq 8 \checkmark$

Condition satisfied!

2 Create Virtual Machines

$G = M1 + M2 + M3$

$H = M2 + M3 + M4$

Job	1	2	3	4	5
G	$11+4+6=21$	$13+3+7=23$	$9+5+5=19$	$16+2+8=26$	$17+6+4=27$
H	$4+6+15=25$	$3+7+8=18$	$5+5+13=23$	$2+8+9=19$	$6+4+11=21$

3 Apply Johnson's Rule

Min = 18 (J2 on H) → J2 LAST

Min = 19 (J3 on G, J4 on H) → J3 FIRST, J4 before J2

Min = 21 (J1 on G, J5 on H) → J1 second, J5 before J4

Sequence: J3 → J1 → J5 → J4 → J2

Answer

Optimal Sequence: J3 → J1 → J5 → J4 → J2

6 Statistical Quality Control - Formulas

Control Limits (3-Sigma)

$$UCL = \text{Mean} + 3\sigma \text{ (Upper Control Limit)}$$

$$CL = \text{Mean (Centre Line)}$$

$$LCL = \text{Mean} - 3\sigma \text{ (Lower Control Limit)}$$

99.73% of data falls within 3 standard deviations from mean in a normal distribution.

X-bar Chart (Mean Chart)

$\bar{X}$ (Grand Mean) = Sum of sample means / Number of samples

$$UCL = \bar{X} + A_2 \bar{R}$$

$$LCL = \bar{X} - A_2 \bar{R}$$

A_2 is a factor depending on sample size (from tables):

n=2: $A_2=1.880$, n=3: $A_2=1.023$, n=4: $A_2=0.729$, n=5: $A_2=0.577$

R Chart (Range Chart)

$\bar{R}$ (Average Range) = Sum of sample ranges / Number of samples

$$UCL = D_4 \times \bar{R}$$

$$LCL = D_3 \times \bar{R}$$

D₃ and D₄ are factors depending on sample size:

n=2: D₃=0, D₄=3.267 | n=3: D₃=0, D₄=2.575

n=4: D₃=0, D₄=2.282 | n=5: D₃=0, D₄=2.115

p-Chart (Proportion Defective)

$$\bar{p} = \text{Total defectives} / \text{Total inspected}$$

$$UCL = \bar{p} + 3\sqrt{(\bar{p}(1-\bar{p})/n)}$$

$$LCL = \bar{p} - 3\sqrt{(\bar{p}(1-\bar{p})/n)}$$

If LCL is negative, use LCL = 0

n = sample size

c-Chart (Defects per Unit)

$$\bar{c} = \text{Total defects} / \text{Number of units inspected}$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}}$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}}$$

Used for counting defects in a constant sample size.

If LCL is negative, use LCL = 0

Process Capability

$$C_p = (USL - LSL) / 6\sigma$$

$$C_{pk} = \min[(USL - \mu)/3\sigma, (\mu - LSL)/3\sigma]$$

USL = Upper Specification Limit

LSL = Lower Specification Limit

Cp ≥ 1 means process is capable

Cpk ≥ 1.33 is generally desired

7 Theory of Constraints - Throughput Accounting

Core TOC Measures

Throughput (T) = Revenue - Truly Variable Costs

Investment (I) = Money tied up in system

Operating Expense (OE) = Money to turn I into T

Derived Measures

Net Profit = Throughput - Operating Expenses

$$NP = T - OE$$

Return on Investment

$$ROI = \text{Net Profit} / \text{Investment}$$

$$ROI = (T - OE) / I$$

Productivity

$$\text{Productivity} = \text{Throughput} / \text{Operating Expenses}$$

$$\text{Productivity} = T / OE$$

Investment Turns

$$\text{Investment Turns} = \text{Throughput} / \text{Investment}$$

$$\text{Investment Turns} = T / I$$

TOC Decision Priority (In Order):

1. Increase Throughput (highest priority - unlimited potential)
2. Reduce Investment
3. Reduce Operating Expenses (limited - can't go below zero)

8

Demand Forecasting - Formulas & Numericals

Simple Moving Average

$$MA_n = (D_1 + D_2 + \dots + D_n) / n$$

Average of last n periods' actual demand.
All periods have equal weight.

Weighted Moving Average

$$WMA = (w_1D_1 + w_2D_2 + \dots + w_nD_n) / (w_1 + w_2 + \dots + w_n)$$

Recent periods typically get higher weights.
Sum of weights should equal 1 (or 100%).

Exponential Smoothing

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

OR

$$F_t = \alpha A_{t-1} + (1-\alpha)F_{t-1}$$

α (alpha) = Smoothing constant ($0 < \alpha < 1$)

- Low α (0.1-0.3): Stable demand, smooths fluctuations
- High α (0.4-0.5): Volatile demand, responds quickly to changes

Forecast Error Measures

$$\text{Error (e)} = \text{Actual} - \text{Forecast} = A - F$$

$$\text{MAD} = \Sigma |A - F| / n$$

$$\text{MSE} = \Sigma (A - F)^2 / n$$

$$\text{MAPE} = (\Sigma |A - F| / A \times 100) / n$$

MAD = Mean Absolute Deviation

MSE = Mean Squared Error

MAPE = Mean Absolute Percentage Error

Problem 1 Exponential Smoothing Forecast**Problem Statement:**

Given the following demand data, calculate forecast for period 6 using exponential smoothing with $\alpha = 0.3$. Initial forecast $F_1 = 100$.

Given Data:

Period	1	2	3	4	5
Actual Demand	110	105	115	120	108

$\alpha = 0.3$, $F_1 = 100$

Solution:

Using: $F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$

1 Calculate Forecasts Step by Step

$$F_1 = 100 \text{ (given)}$$

$$F_2 = F_1 + 0.3(A_1 - F_1)$$

$$F_2 = 100 + 0.3(110 - 100) = 100 + 3 = 103$$

$$F_3 = F_2 + 0.3(A_2 - F_2)$$

$$F_3 = 103 + 0.3(105 - 103) = 103 + 0.6 = 103.6$$

$$F_4 = F_3 + 0.3(A_3 - F_3)$$

$$F_4 = 103.6 + 0.3(115 - 103.6) = 103.6 + 3.42 = 107.02$$

$$F_5 = F_4 + 0.3(A_4 - F_4)$$

$$F_5 = 107.02 + 0.3(120 - 107.02) = 107.02 + 3.89 = 110.91$$

$$F_6 = F_5 + 0.3(A_5 - F_5)$$

$$F_6 = 110.91 + 0.3(108 - 110.91) = 110.91 - 0.87 = 110.04$$

Answer

Forecast for Period 6 (F_6) = 110.04 $\approx$ 110 units

Problem 2 Moving Average and MAD Calculation**Problem Statement:**

Calculate 3-month moving average forecast and MAD for the following data:

Given Data:

Month	1	2	3	4	5	6
Demand	200	220	180	240	210	230

Solution:**1 Calculate 3-Month Moving Average**

$$F_4 = (D_1 + D_2 + D_3)/3 = (200 + 220 + 180)/3 = 600/3 = 200$$

$$F_5 = (D_2 + D_3 + D_4)/3 = (220 + 180 + 240)/3 = 640/3 = 213.33$$

$$F_6 = (D_3 + D_4 + D_5)/3 = (180 + 240 + 210)/3 = 630/3 = 210$$

2 Calculate Errors and MAD

Month	Actual (A)	Forecast (F)	Error (A-F)	Error
4	240	200	40	40
5	210	213.33	-3.33	3.33
6	230	210	20	20
Total Error				63.33

$$\text{MAD} = \text{Total |Error|} / n = 63.33 / 3 = 21.11$$

Answer

$$\text{Forecast for Month 7} = (240 + 210 + 230)/3 = 226.67$$

$$\text{MAD} = 21.11 \text{ units}$$

*

Quick Reference - All Formulas at a Glance

Break-Even Point

$$\text{BEP} = F / (S - V)$$

Centre of Gravity

$$C = \Sigma(d \times W) / \Sigma W$$

Basic Time

$$\text{BT} = \text{OT} \times (\text{Rating}/100)$$

Standard Time

$$\text{ST} = \text{BT} \times (1 + \text{Allow}\%)$$

Value

$$\text{Value} = \text{Function} / \text{Cost}$$

EOQ

$$\text{EOQ} = \sqrt{(2\text{D}\text{C}_o/\text{C}_c)}$$

Total Inventory Cost

$$\text{TC} = (\text{C}_o \times \text{D}/\text{Q}) + (\text{C}_c \times \text{Q}/2)$$

Control Limits

$$\text{UCL/LCL} = \mu \pm 3\sigma$$

Net Profit (TOC)

$$\text{NP} = \text{T} - \text{OE}$$

ROI (TOC)

Exponential Smoothing

MAD

$$ROI = (T - OE) / I$$

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$$

$$MAD = \Sigma |A - F| / n$$

Johnson's Rule Quick Reference

2 Machines (A → B)

Min time on A → Job goes FIRST
Min time on B → Job goes LAST

3 Machines (A → B → C) - Condition Check

If $\text{Min}(A) \geq \text{Max}(B)$ OR $\text{Min}(C) \geq \text{Max}(B)$:
Create $G = A + B$, $H = B + C$
Apply 2-machine rule on G and H

Variable Definitions

Symbol	Meaning
--------	---------

D	Annual Demand
Co	Ordering Cost per order
Cc	Carrying Cost per unit per year
Q	Order Quantity
F	Fixed Cost / Forecast
V	Variable Cost per unit
S	Selling Price per unit
OT	Observed Time
BT	Basic Time / Normal Time
ST	Standard Time
α	Smoothing constant (0-1)
T	Throughput
I	Investment
OE	Operating Expense

